

Introduction:

Let $y = f(x)$, $x_0 \leq x \leq x_n$

Assume that $f(x)$ is single valued and continuous function and that it is known explicitly, then the values of $f(x)$ corresponding to certain given values of x , say $x_0, x_1, \dots, x_n$ can easily be computed and tabulated.

Now consider the converse of above statement, i.e., Given the set of tabular points $(x_0, y_0), \dots, (x_n, y_n)$ satisfying the relation $y = f(x)$ where the explicit nature of $f(x)$ is not known. It is required to find a simpler function, say $\phi(x)$ such that $f(x)$ and $\phi(x)$ agree at the tabulated points. Such a process is called Interpolation. If $\phi(x)$ is a polynomial, then the process is called polynomial Interpolation and $\phi(x)$ is called the interpolating polynomial.

Remark: Different types of Interpolation arise depending on whether $\phi(x)$ is a finite trigonometric series, series of Bessel functions, etc.

Theorem (Weierstrass) If $f(x)$ is continuous in $x_0 \leq x \leq x_n$, then given any $\epsilon > 0$, $\exists$ a polynomial $P(x)$ such that

$$|f(x) - P(x)| < \epsilon \quad \forall x \text{ in } (x_0, x_n).$$

Polynomial Interpolation:

Let $f(x)$ be a continuous function in $[a, b]$ and defined at $n+1$ distinct points $x_0, x_1, \dots, x_n$ such that

$a = x_0 < x_1 < x_2 < \dots < x_n = b$. These points may be equispaced, i.e., $x_{i+1} - x_i = h$, $i = 0, 1, \dots, n-1$ or non-equispaced. The problem of polynomial interpolation is to find a polynomial $p_n(x)$ of degree $\leq n$ which fits the given distinct data exactly, i.e.,

$$p_n(x_i) = f(x_i), \quad i = 0, 1, \dots, n. \quad \text{--- (1)}$$

Such a polynomial is called the Interpolating polynomial.

The conditions (1) are called Interpolating conditions.

The points $x_0, x_1, \dots, x_n$ are called nodal points, arguments or tabular points. If $p_n(x)$ is determined then for any non-tabular point $x = x^* \neq x_i$, we predict that $f(x^*) \approx p_n(x^*)$

Remark 1. Through $n+1$ distinct points, a polynomial of degree $\leq n$ can always be fitted.

2. The polynomial $p_n(x)$ obtained using condition (1) is unique, i.e., for a given data, the interpolating polynomials obtained in two different ways may look different in the form but are identical otherwise.

e.g. $3x^2 + 2x + 5$

+ $10 + 8(x-1) + 3(x-1)^2$ are same poly.

(2)

Lagrange Interpolation: Let $y = f(x)$ be

continuous and differentiable at $n+1$ points in the interval (a, b) . Given $(n+1)$ points

x	x_0	x_1	$\dots$	x_n
$f(x)$	f_0	f_1	$\dots$	f_n

the values of x need not be equally spaced.

We have to find a polynomial of degree n say

$P_n(x)$ s.t

$$P_n(x_i) = f(x_i), \quad i = 0, 1, 2, \dots, n.$$

(Before deriving the General formula, First consider a simpler case, viz., the equation of a straight line passing through two points (x_0, y_0) and (x_1, y_1) . $\therefore$)

$$P_1(x) = \frac{x-x_1}{x_0-x_1} f_0 + \frac{x-x_0}{x_1-x_0} f_1$$

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$y = y_1 + \frac{y_2 - y_1}{x_2 - x_1} (x - x_1)$$

$$= \frac{x - x_1}{x_2 - x_1} y_2$$

$$+ y_1 \left[1 - \frac{x - x_1}{x_2 - x_1} \right]$$

$$= \frac{x - x_1}{x_2 - x_1} y_2 + y_1 \frac{x_2 - x_1 - (x - x_1)}{x_2 - x_1}$$

$$y - y_0 = \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$$

$$y = y_0 + \frac{y_1 - y_0}{x_1 - x_0} (x - x_0)$$

$$= y_0 \left(1 - \frac{x - x_0}{x_1 - x_0} \right) + y_1 \frac{x - x_0}{x_1 - x_0}$$

$$= y_0 \frac{x_1 - x}{x_1 - x_0} + y_1 \frac{x - x_0}{x_1 - x_0}$$

$$\text{where } l_0(x) = \frac{x - x_1}{x_0 - x_1}$$

$$= l_0(x) f_0 + l_1(x) f_1 = \sum_{i=0}^1 l_i(x) f_i \quad \text{--- (1)}$$

$$\text{and } l_1(x) = \frac{x - x_0}{x_1 - x_0}$$

It is clear that

$$l_0(x_0) = 1, \quad l_0(x_1) = 0, \quad l_1(x_0) = 0, \quad l_1(x_1) = 1$$

$$\therefore l_i(x_j) = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{--- (2)}$$

$$\text{Also } \sum_{i=0}^1 l_i(x) = l_0(x) + l_1(x) = 1 \quad \text{--- (3)}$$

Eq (1) is the Lagrange polynomial of degree one passing through two points (x_0, y_0) and (x_1, y_1) called **Linear Interpol.**

In the similar way, the Lagrange polynomial of degree two passing through three points (x_0, y_0) , (x_1, y_1) & (x_2, y_2) is

$$P_2(x) = \sum_{i=0}^2 l_i(x) y_i$$

(**Quadratic Interpolation**)

$$= l_0(x) y_0 + l_1(x) y_1 + l_2(x) y_2$$

$$l_0(x) = \frac{(x-x_1)(x-x_2)}{(x_0-x_1)(x_0-x_2)}$$

$$l_1(x) = \frac{(x-x_0)(x-x_2)}{(x_1-x_0)(x_1-x_2)}$$

$$l_2(x) = \frac{(x-x_0)(x-x_1)}{(x_2-x_0)(x_2-x_1)}$$

where, the $l_i(x)$ satisfy the conditions (2) and (3).

Now $P_n(x)$

$$P_n(x) = \sum_{i=0}^n l_i(x) y_i$$

$$l_i(x) = \frac{(x-x_0)(x-x_1)\dots(x-x_{i-1})(x-x_{i+1})\dots(x-x_n)}{(x_i-x_0)(x_i-x_1)\dots(x_i-x_{i-1})(x_i-x_{i+1})\dots(x_i-x_n)}$$

which satisfy the conditions

$$l_i(x_j) = \begin{cases} 1 & \text{if } i=j \\ 0 & \text{if } i \neq j \end{cases} \quad \text{and} \quad \sum_{i=0}^n l_i(x) = 1$$

Exc: Find Interpolating polynomial for the data (3)

x	-2	-1	1	3
$f(x)$	-15	-4	0	20

$y = 19$
 $x = 3.1196$

There are four points $\Rightarrow$ we will get a polynomial of degree ≤ 3 .

$$P_3(x) = L_0(x)f_0 + L_1(x)f_1 + L_2(x)f_2 + L_3(x)f_3$$

$$x_0 = -2$$

$$f_0 = -15$$

$$x_1 = -1$$

$$f_1 = -4$$

$$x_2 = 1$$

$$f_2 = 0$$

$$x_3 = 3$$

$$f_3 = 20$$

$$L_0(x) = \frac{(x-x_1)(x-x_2)(x-x_3)}{(x_0-x_1)(x_0-x_2)(x_0-x_3)} = \frac{(x+1)(x-1)(x-3)}{(-1)(-3)(-5)}$$

$$= -\frac{1}{15}(x^3 - 3x^2 - x + 3)$$

$$L_1(x) = \frac{(x+2)(x-1)(x-3)}{(1)(-2)(-4)} = \frac{1}{8}(x^3 - 2x^2 - 5x + 6)$$

$$L_2(x) = \frac{(x+2)(x+1)(x-3)}{3(2)(-2)} = -\frac{1}{12}(x^3 - 7x - 6)$$

$$L_3(x) = \frac{(x+2)(x+1)(x-1)}{5 \cdot 4 \cdot 2} = \frac{1}{40}(x^3 + 2x^2 - x - 2)$$

$$P_3(x) = L_0(x) \cdot (-15) + L_1(x) \cdot (-4) + L_2(x) \cdot 0 + L_3(x) \cdot 20$$

$$= -15 \cdot \frac{1}{15} (x^3 - 3x^2 - x + 3) - 4 \cdot \frac{1}{8} (x^3 - 2x^2 - 5x + 6) + 0$$

$$+ 20 \cdot \frac{1}{40} (x^3 + 2x^2 - x - 2)$$

$$= \cancel{x^3 + 6x^2 + 2x - 3} \\ x^3 - x^2 + x - 1$$

Exc: Given $f(2) = 4$, $f(2.5) = 5.5$. Find
Lagrange Interpolating polynomial. Find $P_1(2.2) \approx f(2.2)$

$$x_0 \quad 2 \quad f_0 = 4$$

$$x_1 \quad 2.5 \quad f_1 = 5.5$$

$$P_1(x) = l_0(x) f_0 + l_1(x) f_1$$

$$l_0(x) = \frac{x - x_1}{x_0 - x_1} = \frac{x - 2.5}{-0.5}$$

$$l_1(x) = \frac{x - x_0}{x_1 - x_0} = \frac{x - 2}{0.5}$$

$$P_1(x) = \frac{x - 2.5}{-0.5} \times 4 + \frac{x - 2}{0.5} \times 5.5$$

$$= -8(x - 2.5) + 11(x - 2)$$

$$= -8x + 20 + 11x - 22$$

$$= 3x - 2$$

$$P_1(2.2) = 3 \cdot 2.2 - 2 = 4.6$$

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Inverse Interpolation: Let $y = f(x)$ be some unknown function of independent variable x , whose numerical values for different values of x , say, $x_0, x_1, \dots, x_n$ are given as $y_0 = f_0, y_1 = f_1, \dots, y_n = f_n$. The process of finding the values of the argument x corresponding to a given value of the function y lying b/w two tabulated values with the help of given set of observations is known as Inverse Interpolation. It can be solved by Lagrange Interpolation. Replace x by y .
 Ex: obtain a value of x when $y = 19$

x	0	1	2
y	0	1	20

$$x_0 = 0, x_1 = 1, x_2 = 2$$

$$y_0 = 0, y_1 = 1, y_2 = 20, y = 19$$

$$x = \frac{(y - y_1)(y - y_2)}{(y_0 - y_1)(y_0 - y_2)} \cdot x_0 + \frac{(y - y_0)(y - y_2)}{(y_1 - y_0)(y_1 - y_2)} \cdot x_1 +$$

$$\frac{(y - y_0)(y - y_1)}{(y_2 - y_0)(y_2 - y_1)} \cdot x_2$$

$$= \frac{(19 - 1)(19 - 20)}{(0 - 1)(0 - 20)} \cdot 0 + \frac{(19 - 0)(19 - 20)}{(1 - 0)(1 - 20)} \cdot 1 +$$

$$\frac{(19 - 0)(19 - 1)}{(20 - 0)(20 - 1)} \cdot 2 = 2.8$$

Error of Interpolation: $f(x)$ has continuous derivative of order upto $n+1$ $\forall x \in [a, b]$. Since $f(x)$ is approximated by $P_n(x)$, the computed results contain errors. The difference $f(x) - P_n(x)$ is called the error of interpolation or truncation error. We write

$$E_n(f, x) = f(x) - P_n(x)$$

obviously, at the tabular points x_i , $E_n(f, x_i) = 0$, $i = 0, 1, 2, \dots, n$.

and $E_n(f, x^*) \neq 0$ for $x^* \neq x_i$.

consider an auxiliary function

$$\phi(t) = f(t) - P_n(t) - \underbrace{\left[f(x) - P_n(x) \right]}_{\text{①}} \frac{(t-x_0) \dots (t-x_n)}{(x-x_0) \dots (x-x_n)}$$

We observe that $\phi(t) = 0$ at $t = x$ and $t = x_i$ for $i = 0, 1, 2, \dots, n$.

$\therefore \phi(t)$ has $n+2$ zeros. Apply Rolle's Theorem repeatedly for $\phi(t)$, $\phi'(t)$, $\dots$, $\phi^n(t)$, we obtain

$$\phi^{n+1}(\xi) = 0, \text{ where}$$

$$\min(x_0, x_1, \dots, x_n) < \xi < \max(x_0, x_1, \dots, x_n)$$

Differentiating ① w.r.t $n+1$ times, we get

$$\phi^{n+1}(t) = f^{n+1}(t) - \frac{[n+1] \left[f(x) - P_n(x) \right]}{(x-x_0) \dots (x-x_n)}, \quad d(P_n) = n$$

setting $\phi^{n+1}(\xi) = 0$, we get

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$$f^{n+1}(\xi) - \frac{|n+1| (f(x) - p_n(x))}{(x-x_0) \cdots (x-x_n)} = 0$$

$$\Rightarrow E_n(f, x) = f(x) - p_n(x) = \frac{1}{|n+1|} \left[(x-x_0) \cdots (x-x_n) \right] f^{n+1}(\xi)$$

The bound for the error is given by

$$|E_n(f, x)| \leq \frac{1}{|n+1|} \left[\max_{a \leq x \leq b} |(x-x_0) \cdots (x-x_n)| \right] M_{n+1}$$

$$M_{n+1} = \max_{a \leq x \leq b} |f^{n+1}(x)|$$